

Abstract No.

Role of ZBTB18 in O-GlcNAcylation in glioblastoma

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Glioblastoma (GBM) is the most aggressive primary brain tumor. We have identified Zinc finger and BTB domain-containing protein 18 (ZBTB18) as a potential glioma suppressor whose downregulation is related to poor patients' outcome. Our preliminary metabolomics analysis, further confirmed by western blot, indicates that the re-expression of ZBTB18 in GBM cell lines reduces the amount of O-linked N-acetylglucosamine (O-GlcNAc), responsible for the O-GlcNAcylation process, a post-translational modification that involves the attachment of O-GlcNAc moieties to serine and threonine of proteins, and that could promote tumorigenesis by coordinating glycolysis and the TCA cycle. Aberrant O-GlcNAcylation might contribute to metabolic disorders, such as insulin resistance, suggesting it could also play a role in the altered metabolism occurring in cancer cells. In the brain, this reversible and dynamic glycosylation is involved in essential processes, such as synaptic activity and neuronal morphogenesis.

Interestingly, GFPT2, a key limiting enzyme involved in the production of O-GlcNAc, belongs to the group of mesenchymal signature genes in GBM and it is regulated by ZBTB18, as confirmed by qRT-PCR analysis. O-GlcNAcylation is controlled by two enzymes, O-GlcNAc transferase (OGT) and O-GlcNAcase (OGA). Therefore, we sought to identify proteins modified upon ZBTB18 re-expression using the PTMScan® O-GlcNAc [GlcNAc-S/T] Motif Kit (Cell Signaling), followed by mass spectrometry. We identified a few candidates showing increased or reduced O-GlcNAcylation upon exposure to TMG, an OGA inhibitor, or ZBTB18 re-expression, respectively. These pieces of evidence suggest that O-GlcNAcylation may play an important part in ZBTB18 tumor repressive function.



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